“AI GENERATED TEXT DETECTION USING RoBERTa”

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Problem Statement & Solution

Problem

With the rapid growth of AI tools, distinguishing between **human-written** and **AI-generated text** has become a major challenge in academics, research, and content verification.

Solution

Developed an **AI-based text detection system** using the **RoBERTa model** to accurately classify text as human-written or AI-generated based on language patterns and context.

Why Do I Use RoBERTa Model ?

- Transformer-based advanced NLP model
- Strong understanding of context and sentence structure
- Detects subtle differences in writing style
- Provides higher accuracy than traditional models
- Efficient for text classification tasks

Implementation Workflow

Dataset Collection

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Text Preprocessing

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Tokenization

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RoBERTa Model Loading

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Model Training

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Model Evaluation

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Prediction

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Result Visualization

Model Working

- Input text is converted into tokens using tokenizer
- Tokens are transformed into numerical representations
- RoBERTa processes the input and learns language patterns
- Model classifies text as:

  Human-written

  AI-generated

```
Running formal evaluation on the 20% unseen test dataset...
This will take about 1-2 minutes...

==================================================
              CLASSIFICATION REPORT
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                  precision    recall  f1-score   support

Human (Class 0)     0.9999    0.9947    0.9973      9683
   AI (Class 1)     0.9947    0.9999    0.9973      9593

       accuracy                         0.9973     19276
      macro avg     0.9973    0.9973    0.9973     19276
   weighted avg     0.9973    0.9973    0.9973     19276


Generating Confusion Matrix Plot...
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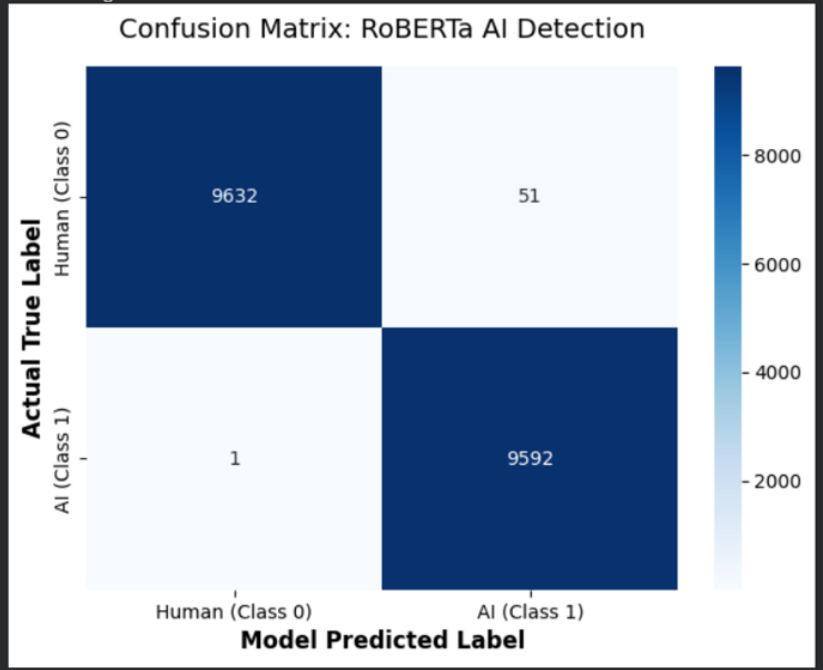

Results Comparison (With Reference Paper)

Evaluation Metrics	Reference Baseline (Guo et al., 2023)	My Project (Optimized RoBERTa)
Model Architecture	RoBERTa-Base	RoBERTa-Based (Fine-tuned)
Dataset Size	~48000 QA pairs	~96000 text samples
Overall Accuracy	99.0%	99.73%
F1-Score	98.8%	99.73%
Precision (Human)	99.0%	99.99%
Precision (AI Class)	98.0%	99.47%
Adversarial Stress Testing	Not Evaluated	Evaluated (Paraphrasing & Formality)

Conclusion

1. The AI text detection system was developed using the RoBERTa model and trained on a large combined dataset to effectively learn patterns distinguishing human-written and AI-generated text. The experimental results achieved an accuracy of 99.73% along with a high F1-score, demonstrating the model's strong detection capability.
2. Previous research primarily focused on classifying text based only on direct human and AI-generated inputs. However, in this project, an additional stress testing approach was introduced by applying paraphrasing and modifying text styles. This tested the robustness of the model in more realistic scenarios where AI-generated content may be altered.

Thank you!